

Machine Learning Project Report

Project Title: Predicting Teenage Depression Risk Using Machine Learning

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Course: Machine Learning

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Abstract

This project focuses on predicting teenage depression risk using machine learning techniques. The dataset includes features such as sleep hours, daily social media usage, academic performance, stress levels, and anxiety levels. Various classification models were applied, and their performance was evaluated. The goal is to identify the most accurate model for mental health screening and to understand key factors influencing depression risk in teenagers.

1. Introduction

Machine Learning is widely used in predictive analytics, and its application in healthcare and mental health is growing rapidly. Predicting depression risk helps school counselors, healthcare providers, and parents make informed decisions and intervene early.

- **Problem Statement:** To build a model that predicts a teenager's depression risk (Low, Medium, High) based on given features like lifestyle habits and stress indicators.
- **Objectives:**
 - Understand dataset features
 - Perform data preprocessing
 - Perform EDA
 - Apply ML models
 - Evaluate performance

2. Dataset Description

- **Source:** Kaggle dataset (Teen_Mental_Health_Dataset.csv)
- **Number of rows:** 2500 (80% training / 20% testing split)
- **Features:**

Feature	Description
age	
gender	
daily_social_media_hours	Hours spent on social media daily
platform_usage	
sleep_hours	Hours of sleep per night
Screen_time_before_sleep	
academic_performance	
physical_activity	
social_interaction_level	
stress_level	Self-reported scale (1-10)
anxiety_level	
depression_risk	

3. Data Preprocessing

- **Steps performed:**
 - Loaded and understood the dataset
 - Handling missing values (None found, Verified clean data)
 - Encoding categorical variables (Ordinal encoding for risk, One-Hot encoding for gender/platform)
 - Split dataset into training and testing sets

4. Exploratory Data Analysis (EDA)

- **Distribution plots** (Analyzed age distribution across risk categories)
- **Correlation matrix** / Scatter plots (Visualized Sleep vs. Social Media)
- **Feature importance visualization** (Extracted from Random Forest)
- **Key Insights:**
 - Stress and Anxiety levels strongly influence depression risk.
 - High social media hours combined with low sleep hours play a major role in classifying High-Risk teens.

5. Model Building

- **Algorithms Used:**
 - Logistic Regression (with balanced class weights)
 - Decision Tree
 - Random Forest (Base and Hyperparameter Tuned via GridSearchCV)

Example Code (Python)

Python

```

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

X = data.drop('depression_risk', axis=1)
y = data['depression_risk']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = RandomForestClassifier(n_estimators=200, max_depth=10,
random_state=42)
model.fit(X_train, y_train)

```

6. Model Evaluation

- **Metrics Used:**
 - Accuracy
 - Precision, Recall, and F1-Score (via Classification Report)
- **Results:**

Model	Accuracy	High-Risk Recall
Logistic Regression (Base)	76.00%	77.00%
Logistic Regression (Balanced)	75.60%	79.00%
Decision Tree	73.20%	70.00%
Random Forest (Tuned)	77.60%	97.00%

7. Results & Discussion

- Random Forest performed best, ultimately achieving an accuracy of 77.60% after grid search optimization.

- The tuned Random Forest model achieved an outstanding 97% Recall for the "High Risk" category, meaning it successfully minimizes dangerous false negatives.
- Overfitting observed in Decision Tree, which had the lowest overall accuracy.
- All models struggled to perfectly classify the "Medium" risk group due to overlapping feature distributions.

8. Conclusion

The project successfully built a predictive model for teenage depression risk. Random Forest gave the best results, heavily relying on stress, anxiety, and social media hours to make its predictions. Future improvements can include deep learning models and larger datasets.

9. Future Scope

- Use Neural Networks to capture more complex non-linear relationships.
- Deploy model using web apps for school counselors to use as an interactive dashboard.
- Use real-time data from wearable devices tracking sleep and heart rate.

10. References

- Scikit-learn documentation for classification and hyperparameter tuning.
- Matplotlib and Seaborn documentation for data visualization.
- Research papers on adolescent mental health and screen time.